

Flow Simulation Around Cambered Airfoil by Using Conformal Mapping and Intermediate Domain in Lattice Boltzmann Method

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Abstract In this paper the developed interpolation lattice Boltzmann method is used for simulation of unsteady fluid flow. It combines the desirable features of the lattice Boltzmann and the Joukowski transformation methods. This approach has capability to simulate flow around curved boundary geometries such as airfoils in a body fitted grid system. Simulation of unsteady flow around a cambered airfoil in a non-uniform grid for the first time is considered to show the capability of this method for modeling of fluid flow around complex geometries and complicated long-term periodic flow phenomena. The developed solver is also coupled with a fast adaptive grid generator. In addition, the new approach retains all the advantages of the standard lattice Boltzmann method. The Strouhal number, the pressure, the drag and the lift coefficients obtained from the simulations agree well with classical computational fluid dynamics simulations. Numerical studies for various test cases illustrate the strength of this new approach.

Keywords LBM · Incompressible flow · Conformal mapping · Non-uniform grid

1 Introduction

Recently, lattice Boltzmann method (LBM) has become one of the attractive computational approaches for fluid flow calculations. Originally, LBM was formulated on isotropic grid systems such as orthogonal grid with equal spacing. In fact the LBM with BGK collision models is still plagued with restriction of lattice-uniformity in physical space, which makes the scheme macroscopically similar to a uniform Cartesian-grid solver. From any practical problems, irregular grids or meshless structures and immersed boundary methods are always preferred due to their simplicity for accurate description of curved boundaries and also their

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computational efficiency. Currently there are three ways to overcome the restriction of the lattice-uniformity [1].

The first is based on the solution of the discrete Boltzmann. The discrete Boltzmann equation can be solved by conventional finite difference schemes (FDLBM) with the use of coordinate transformation [2,3] or by finite volume method (FVLBM) [4,5]. The FDLBM and FVLBM have disadvantages of losing of locality and large numerical diffusion in the region where the flow gradients are large.

The second is the grid refinement technique which was first presented by Filippova and Hänel [6] and further improved by Mei et al. [7]. This technique is based on the conventional adaptive mesh refinement method using a coarse grid in the whole domain and a finer grid in critical regions. Second order interpolation is used on the interfaces and boundaries of different grid levels. However, since the boundaries are not exactly located at the grid nodes in this grid refinement treatment; it is difficult and inconvenient to implement the boundary conditions

The last one is called “Interpolation-Supplemented Lattice Boltzmann Equation” (ISLBE) model that was proposed by He et al. [8]. In this method, interpolation is applied at every time step in order to obtain the density distribution function at the grid points. The method applies interpolation to update the particle distribution functions on the coarse portion of the grid after the propagation step. He et al. [9] applied the ISLBE method to simulate fluid flow in a 2D symmetric channel with a sudden expansion on a non-uniform rectangular grid. Since the grid was coarse in one direction, one-dimensional quadratic interpolation was used to obtain a result with second-order accuracy. Later, He and Doolen [10] constructed a curvilinear coordinate system by applying the ISLBE method to simulate vortex shedding behind a circular cylinder.

Two-dimensional second-order upwind interpolation was applied in their simulation. Comparing with the other non-uniform grid techniques, the ISLBE method is easy for programming, parallel computation, and incorporation of microscopic interactions in multiphase flows computations [11]. It is valid for grids with arbitrary shapes and for both the BGK scheme and the Somers–Eggels scheme. The Somers–Eggels formulation of the LBM model is based on the 4D face-centered-hyper-cubic (FCHC) lattice that in this formulation they Somers and Eggels obtained an expression for the collision operator Ω_i in Eq. (1) by using an asymptotic expression for distribution function [12] The main challenge of the uniform grid is to offer high resolution near a solid body and to place the outer boundary far away from the body. So it is desirable to use body fitted grid system. This study was conducted based on two objectives. One is to derive a new scheme which is free of the lattice uniformity and has good accuracy. The other is to develop a general and efficient LBM solver based on the new scheme.

Based on Navier-Stokes equations, the computation of this unsteady flow case with massive separation demonstrates the ability of the analysis to treat the flow past this Joukowski Airfoil at higher incidence. Moreover, comparing with the standard lattice Boltzmann, the DILBM has better computational stability.

2 Numerical Method

2.1 Lattice Boltzmann Method (LBM)

The LBM utilizes a particle distribution function to describe the collective behavior of fluid molecules. In LB methods, particles are assumed to move synchronously along the bonds

of a regular lattice, and satisfy the discrete form of the lattice Boltzmann equation. For flow field, the evolution equation is given as the following form:

$$f_i(\mathbf{r} + \mathbf{e}_i \delta t, t + \delta t) - f_i(\mathbf{r}, t) = \Omega_i(f) \quad (1)$$

where f_i is the density distribution function, which depends on position $\mathbf{r}$ in the physical space, $\mathbf{e}_i$ is the particle discrete velocity and t is the discrete time, δt is the time step. A simple model for collision operator $\Omega_i(f)$ is BGK model (Bhatnagar et al. 1954) [13, 14] given by:

$$\Omega_i(f) = -\frac{1}{\tau} (f_i(\mathbf{r}, t) - f_i^{eq}(\mathbf{r}, t))$$

τ is the single relaxation parameter related to the hydrodynamic viscosity and f_i^{eq} is the equilibrium density distribution function which depends on the local macroscopic variables and is given as the following form:

$$f_i^{eq} = \rho \cdot w_i \left[1 + \frac{3}{c_s} \vec{e}_i \cdot \vec{u} + \frac{9}{2c_s} (\vec{e}_i \cdot \vec{u})^2 - \frac{3}{2c_s} \vec{u} \cdot \vec{u} \right] \quad (2)$$

where c_i is the particle velocity and w_i is the weighting factor for discretization of the equilibrium distribution function, that is depend on the specific choice of the discrete velocities [15, 16]. The macroscopic density and momentum are defined as particle moments of the distribution function f_i :

$$\begin{aligned} \rho &= \sum_{i=1}^M f_i \\ \rho u &= \sum_{i=1}^M f_i e_i \end{aligned} \quad (3)$$

The equations of state and kinematic viscosity are defined as:

$$\begin{aligned} P &= \rho c_s^2 \\ \nu &= c_s^2 \left(\tau - \frac{1}{2} \right) \delta t \end{aligned} \quad (4)$$

where c_s is the speed of sound.

2.2 Developed Interpolation Lattice Boltzmann Method (DILBM)

The serious limitation of the cartesian grid is that complex, curved geometries cannot be handled without losing boundary integrity. Furthermore, the traditional LB method requires a uniform lattice everywhere, which results in a large number of gridpoints, which in turn, increases the computational cost tremendously [17, 18]. Moreover the interpolation supplemented lattice Boltzmann method (ISLBM) isn't able to simulation of fluid flow around complex geometries. Therefore in the present work it is proposed to extend the traditional LBM and ISLBM to handle curved geometries such as cambered airfoil.

In order to generate a body-fitted grid system around a desired geometry, a new adaptive grid generation technique based on Joukowski transformation is used. In this technique the uniform Cartesian grid system is mapped onto a non-uniform polar grid system around a circular cylinder and then the polar grid is mapped onto an O-type non-uniform body-fitted grid around a cambered airfoil by using a conformal mapping. The conformal mapping is performed through the transformation of grid points from polar coordinates to physical

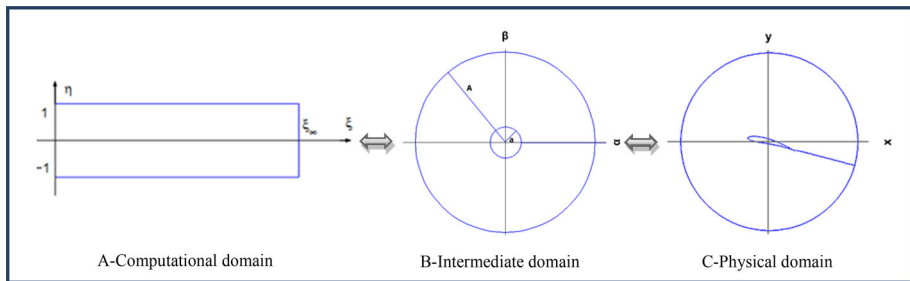


Fig. 1 Domains of the computations

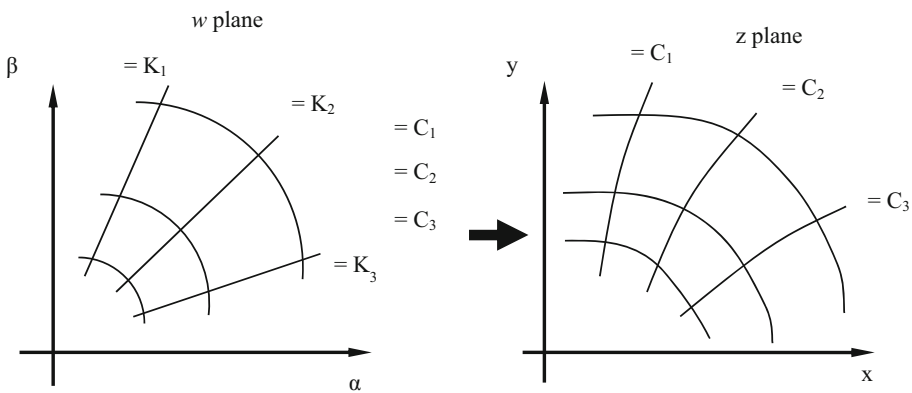


Fig. 2 Joukowski transformation

coordinates. In fact we have three domains: computational domain (ξ, η) plane, intermediate domain (α, β) plane, and physical domains (x, y) plane, these domains are shown in Fig. 1.

In aerodynamics applications the Joukowski transform is the most commonly used transformation [19];

$$z = w + \frac{c^2}{w} \quad (5)$$

Here, c is a constant that shows the circumference of the circle passes through the critical point $w=c$ for the Joukowski transformation, which corresponds to the sharp trailing edge of airfoil in the z plane. Graphically, a conformal mapping transforms a complex plane z ($z = x + iy$) into a complex plane in a new variable w ($w = \alpha + i\beta$) [12] as shown in Fig. 2.

In Fig. 2 a polar grid in w plane is transformed into an equivalent form in z plane using a transform function $z = f(w)$. This property of mapping is implemented for grid generation.

A typical non-uniform grid around a camber airfoil is shown in Fig. 3. It can be seen clearly that the grid points are clustered near the walls to capture the thin boundary layer. The resolution of the grid is 161×191 .

In order to implement the LBM, in the body fitted grid system, pre-collision values for the distribution function $f_i(x_m, y_n, t)$ and equilibrium distribution at each grid point are determined based on the values of the macroscopic quantities, then the collision process is carried out according to Eq. (1) and post-collision values of distribution function $f_i(x_m, y_n, t)$ are obtained then the streaming step starts. At this step, the distribution function for each e_α at its new position $x - e_\alpha \delta t$ is known. In order to determine the new coordinates of the

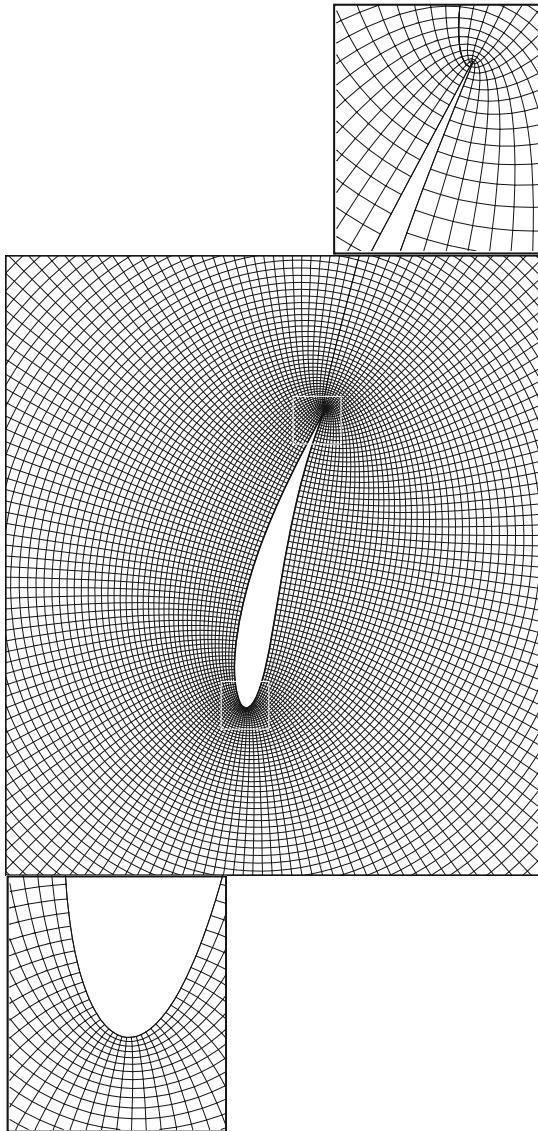


Fig. 3 Body-fitted grid generated around the cambered airfoil, the leading and trailing edges by using the transformation

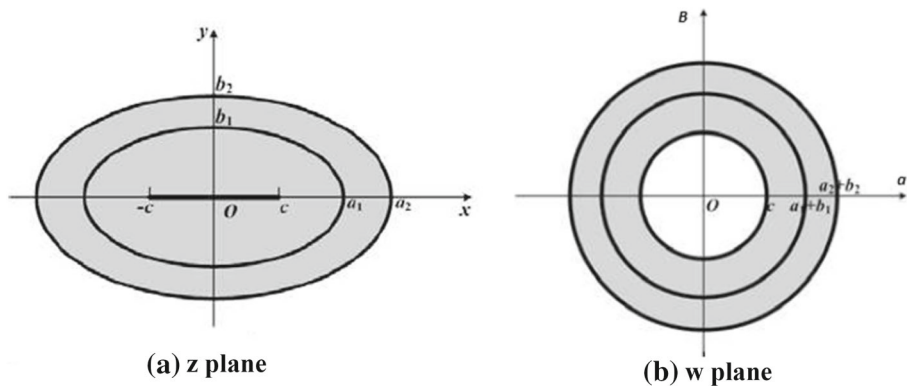


Fig. 4 Inverse Joukowski mapping. **a** The confocal elliptical cluster. **b** Circle cluster with a cut circle of radius c

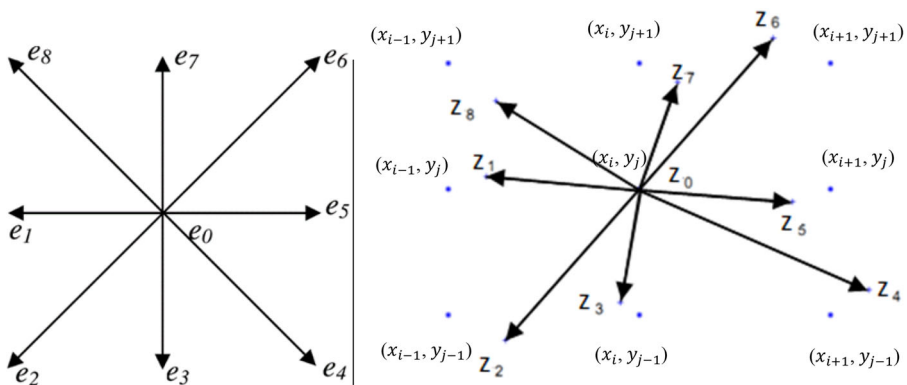


Fig. 5 $e(1,0)$ vectors on computational plane and physical plane

points in the computational domain, an intermediate domain is required. In fact, this domain communicates the positions of the particles in the physical domain and their corresponding positions in the computational domain after streaming step.

Using new positions of the points in the physical domain after the streaming step, their corresponding positions in the intermediate domain are determined by using an inverse of the Joukowski transformation (Fig. 4).

The typical inverse Joukowski transformation maps a family of confocal elliptical in the z -plane with the same focal length onto a family of circles in the w -plane which have a cut circle of radius c .

$$w = \frac{z}{2} \pm \sqrt{\frac{z^2}{4} - c^2} \quad \boxed{z = x + iy} \quad \begin{cases} \text{if } x > 0 & \text{then } w = \frac{z}{2} + \sqrt{\frac{z^2}{4} - c^2} \\ \text{if } x < 0 & \text{then } w = \frac{z}{2} - \sqrt{\frac{z^2}{4} - c^2} \end{cases} \quad (6)$$

After clarifying the new positions of the points in the intermediate domain, their corresponding points in the computational domain can be determined. By using this procedure, we can correlate the body-fitted grid system and the computational grid system (uniform grid).

According to Fig. 5 the values of the distribution functions at the original lattice site (x_m, y_n) at time $t + \delta t$ are unknown after mapping. These values are obtained through interpolation from the values available at the nearest neighbouring nodes, i.e. values of $f_i(x_m, y_n, t + \delta t)$ are computed from the values of $f_i(x_m + \delta t \xi_i, y_m + \delta t \xi_i, t + \delta t)$ based on a suitable interpolation technique. Here, second order accurate Lagrangian interpolation is used as:

$$f_i(m, n) = \sum_{\alpha=0}^2 \sum_{\beta=0}^2 \Gamma_{m,\alpha} \Lambda_{n,\beta} f_i(m - \alpha \times \text{sign}(1, \cos((i-1)\pi/2)), n - \beta \times \text{sign}(1, \sin((i-1)\pi/2))) \quad (7)$$

In writing the above expression, it has been assumed that the f_i 's are stored as arrays and m, n are the array indices. The coefficients $\Gamma_{m,\alpha}$ and $\Lambda_{n,\beta}$ are given as

$$\begin{aligned} \Gamma_{m,0} &= \frac{(x(m) - x(m-1) - \delta x)(x(m) - x(m-2) - \delta x)}{(x(m) - x(m-1))(x(m) - x(m-2))}, \\ \Gamma_{m,2} &= \frac{(x(m) - x(m-1) - \delta x)(-\delta x)}{(x(m) - x(m-2))(x(m) - x(m-2))}, \\ \Gamma_{m,1} &= \frac{(\delta x)(x(m) - x(m-2) - \delta x)}{(x(m) - x(m-1))(x(m) - x(m-2))}, \\ \Gamma_{n,0} &= \frac{(y(m) - y(m-1) - \delta y)(y(m) - y(m-2) - \delta y)}{(y(m) - y(m-1))(y(m) - y(m-2))}, \\ \Lambda_{n,1} &= \frac{(\delta y)(y(n) - y(n-2) - \delta y)}{(y(n) - y(n-1))(y(n-1) - y(n-2))}, \\ \Lambda_{n,2} &= \frac{(y(n) - y(n-1) - \delta y)(-\delta y)}{(y(n-1) - y(n-2))(y(n) - y(n-2))}, \end{aligned}$$

where δx and δy are the lattice spacing and x, y are position of nodes after streaming in the physical domain. The method can be used for resolving flow fields in a Cartesian grid system with clustering in high gradient regions. Here the algebraic equations are replaced by the Joukowski transformation to make connection between grid points in the computational domain and the physical domain in the algebraic grid generation methods. The developed method in this work is based on the fact that the distribution function is a continuous function in the physical space and can be well defined for any grid systems (Fig. 6).

2.3 Boundary Conditions

Boundary conditions are an essential issue in the LBM. In this work a second order half way bounce back scheme which shifts non slip boundary into fluid by one half lattice unit is proposed [20, 21]. So by defining one layer of virtual nodes inside the wall, the possibility of using the second order boundary condition and interpolation is provided. The below condition is applied at each time step for calculating the values of ghost nodes.

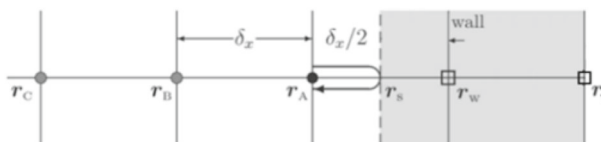


Fig. 6 Illustration of interpolation bounce back boundary condition

$$\left(\frac{\partial^2 f}{\partial h^2}\right)_i = \frac{f_i^A - 2f_i^w + f_i^v}{h^2} + O(h^2) \quad (8)$$

f_i^v , f_i^A are distribution functions on the virtual boundary and the first inner layer of the fluid flow and f_i^w is distribution function on the wall. After this extrapolation solution process is carried out for all nodes.

3 Numerical Simulation

We have proposed a new algorithm to implement the lattice Boltzmann method on arbitrary non-uniform grids. A computer code is developed using numerical methods coming from Poozesh et al. [22]. The overview of the algorithm can be arranged as follow:

- (1) Generation of the Cartesian grid,
- (2) Mapping from Cartesian system to circular system,
- (3) Mapping from circular system to physical system,
- (4) Initialization of the distribution functions,
- (5) Imposing boundary conditions,
- (6) Collision operation which consists in the calculation of f ,
- (7) Propagation operation which consists in the calculation of distribution function at $t + \delta t$,
- (8) Finding new positions of the points in the intermediate domain using inverse of the Joukowski transformation after the streaming step,
- (9) Finding new positions of the points in the Cartesian grid,
- (10) Using a second-order upwind approximation to calculate the distribution function after the streaming step,
- (11) Testing the convergence criterion for velocity in two consecutive time [23], if false return to step (5).

The new algorithm retains all the advantages of the previous lattice Boltzmann method, such as parallelism of the method, ease of the programming, and the capability to incorporate microscopic interactions.

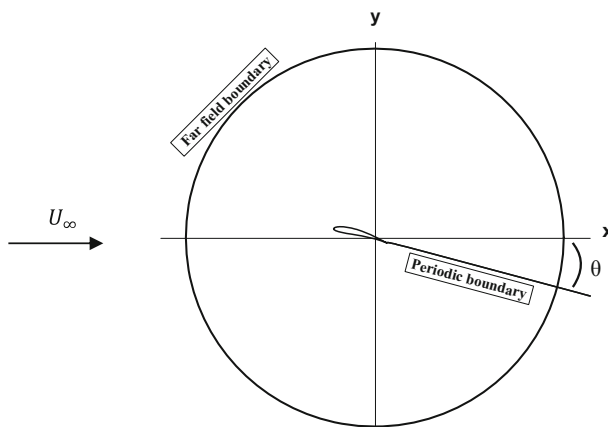


Fig. 7 Schematic diagram of the flow field around airfoil

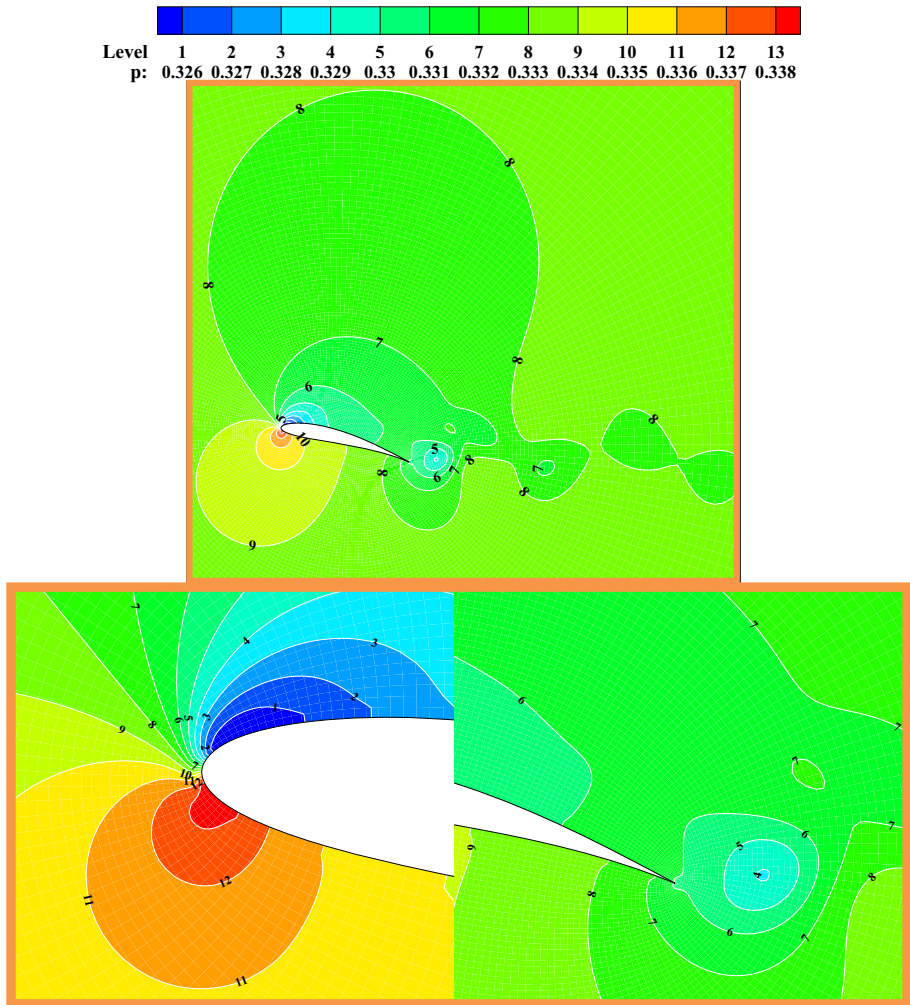


Fig. 8 Pressure contours around camber airfoil at 15° angles of attack

In order to examine the performance of the developed lattice Boltzmann scheme to simulate the incompressible viscous flows on an arbitrary grid, flow around a cambered airfoil was evaluated at various angles of attack (AOA = 0, 2.5, 5, 7.5, 10, 12.5, 15). For all of the test cases the Reynolds number was 1000 and the free stream velocity was 0.1 m/s. The far field pressure was set equal to $1/3$, and the ratio of the streaming length to the radius of a cylinder dx/a was 0.01. The time step, in units of a/U , was 0.001. Figure 7 shows the physical domain and the boundary conditions. The results were compared with those of the conventional CFD calculations [24], which were done by the authors using an in-house developed computer code. Some results of the CFD code were reported in [25]. In the CFD calculations, the grid size of 161×9 with a time step size of 0.001s was fine enough to obtain grid independent results

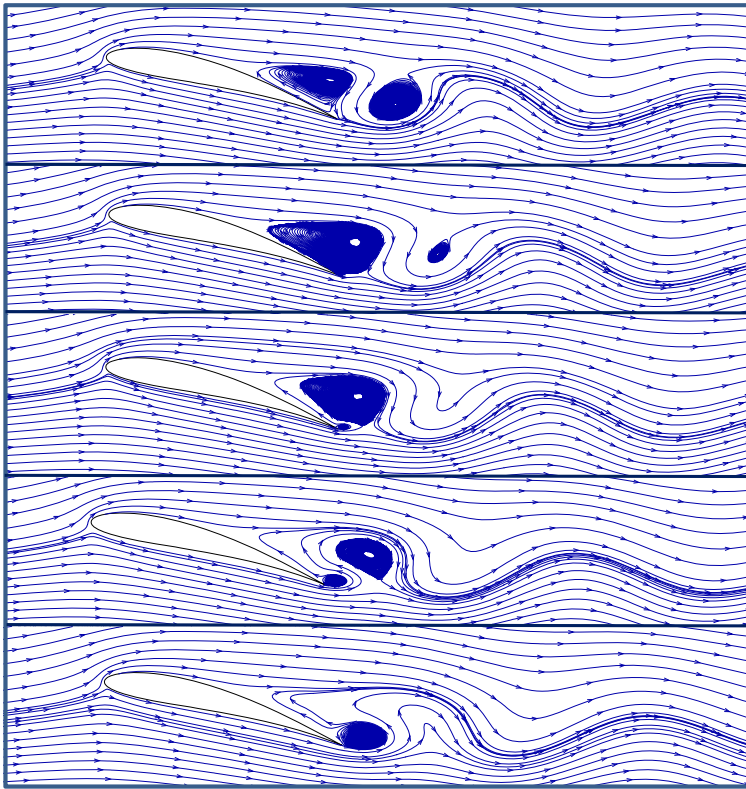


Fig. 9 Sequence of streamlines in a complete shedding cycle, separated by intervals of $T/5$ ($Re=1000$, $AOA=15^\circ$)

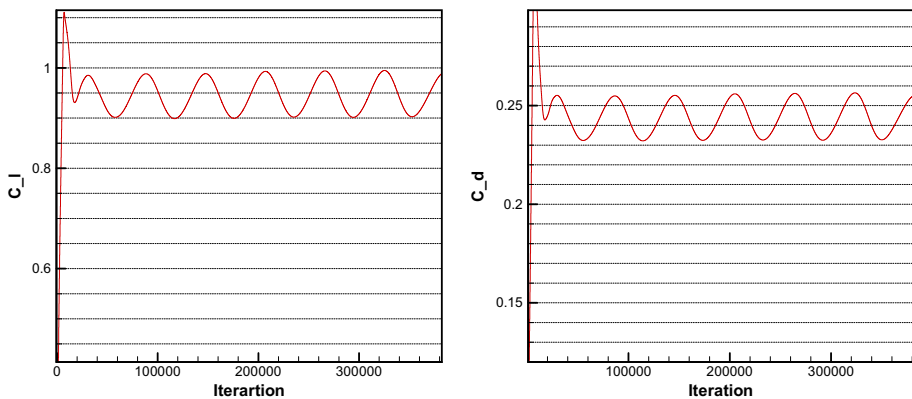


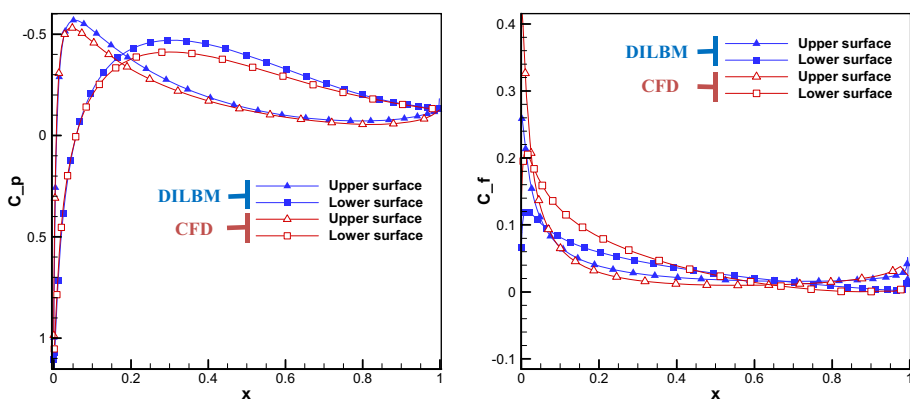
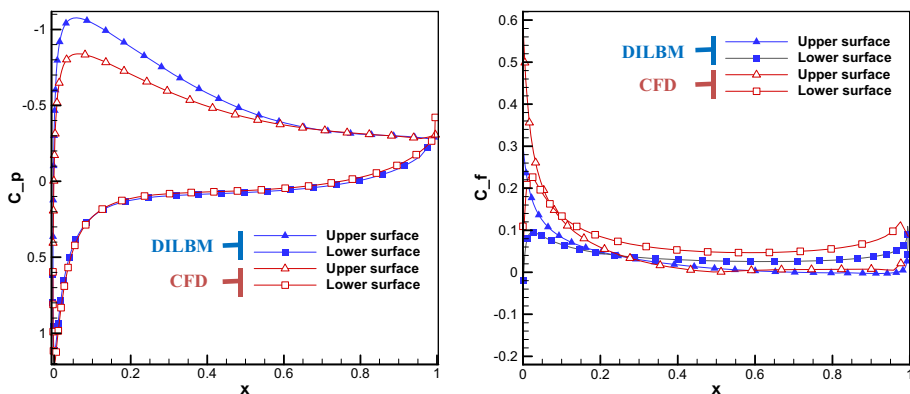
Fig. 10 The drag and lift coefficients for $Re=1000$ and $AOA=15^\circ$

3.1 Results and Discussion

Flow around an unsymmetrical airfoil with a relatively small Reynolds number $Re=1000$ and angle of attack of 15° is simulated to show the ability of the present method. Simulation of

Table 1 Comparison of aerodynamic quantities at 15° angle of attack and $Re = 1000$

Work	C_L	C_D	St
CFD	0.254	0.85	0.69
DILBM	0.245	0.914	0.682

**Fig. 11** Comparison of the pressure and friction coefficients on the cambered airfoil for $Re = 1000$ at 0° angle of attack**Fig. 12** Comparison of the pressure and friction coefficients on the cambered airfoil for $Re = 1000$ at 10° angle of attack

viscous flow over a cambered airfoil predicts vortex shedding with a huge shear layer zone on the suction side of the airfoil. Figure 8 Shows pressure contours around the airfoil. It can be seen that vortices are shedding behind the airfoil.

The global view of the vortex shedding process can be well illustrated using streamline. Figure 9 shows the time evolution of vortex shedding over a complete cycle using a sequence of streamlines separated by intervals of $T/5$, which T is the period of the shedding cycle.

Figure 10 shows variations of drag and lift coefficients with number of iterations. The coefficients vary periodically due to vortex shedding phenomenon [26,27]. Table 1 gives a comparison between the mean values of drag and lift coefficients and also Strohal number

Fig. 13 Distribution of pressure coefficients on the airfoil wall in different angle of attacks

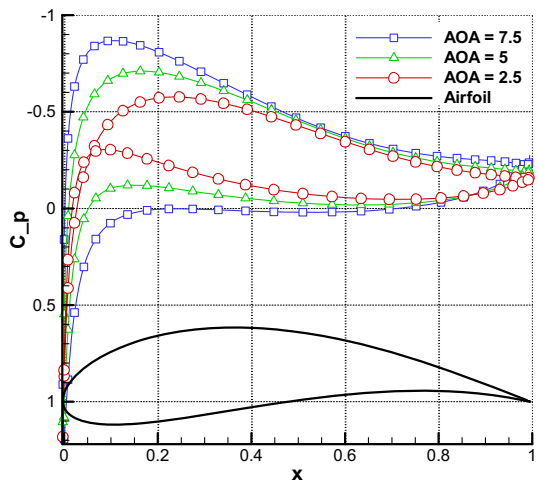


Table 2 Comparison of the quantities at different angle of attack and $Re = 1000$ with CFD results

Angle of Attack	Coefficient	Present work	CFD
0	C_D	0.121	0.121
	C_L	0.067	0.063
5	C_D	0.17	0.184
	C_L	0.39	0.369
10	C_D	0.201	0.212
	C_L	0.61	0.572

calculated by the proposed method with those of CFD calculations. The differences between the results are satisfactory.

Figures 11 and 12 show pressure coefficient and friction factor distributions along the airfoil cord. There is a good agreement between the results of our simulation and those of CFD calculations. The pressure of free stream is assumed $1/3$. Figure 12 gives these distributions for flow with 10° angle of attack.

Pressure distributions for various angles of attack are given in Fig. 13. As the angle of attack increases the difference between the lower and upper surfaces increases this leads to higher lift coefficient. The calculated lift and drag coefficients at various angles of attack are shown in Table 2. In this table the results of the present method and those of CFD are compared. The accuracy of the present results is satisfactory.

4 Conclusions

In this paper, we used the Developed Interpolation Lattice Boltzmann Method (DILBM) as an alternative for other lattice Boltzmann schemes to improve the flexibility of the LBM for fluid flow simulations at low Reynolds numbers and body fitted grid systems. The success of this study shows that the DILBM can be used as a versatile CFD tool with the ability to simulate complicated flows. In particular, the combination of Interpolation Supplemented

Lattice Boltzmann Method (ISLBM) and transformation gives much more flexibility in choice the geometry in body fitted grid system. Moreover this body fitted grid approach avoids the traditionally time consuming non uniform meshing process.

By using the DILBM, the restriction for generation of body fitted grid system is removed and variable grids can be used as desired. Moreover the code generates the mesh automatically by using mapping method.

Developed interpolation method can be used to predict vortex shedding at the behind of vertical plate and symmetrical and non-symmetrical airfoil at high angles of attack. The simplicity of the Cartesian grid for the Eulerian variables facilitates the LBM, and the generality of the curvilinear grid for the Lagrangian variables makes it easy to model the fluid flow problem. It is worth to mention that the proposed method has the advantage of removing the oscillations generated by standard lattice Boltzmann methods on the curved boundaries.

Another advantage of the DILBM is that it does not change the locality of the LB method. Therefore, it is still able to exploit fully the power of parallel computing.

The numerical results were compared with the benchmark data. The present technique is able to apply to the other aerodynamic mappings and also the algebraic equations can be replaced by the Joukowski transformation to make the connection between grid points in the computational domain and the physical domain in the algebraic grid generation methods for more developments of method.

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